

Ibn Tofail University

Rigid Body Mechanics

Problem Set IV

Course Questions*

1) Show that the angular momentum and the dynamic moment are two antisymmetric vector fields.

2) Show that for any point A of the solid S of mass M and center of inertia G

$$\vec{L}_A(S/R_o) = I_A(S)\vec{\Omega}(S/R_o) + M\vec{AG} \times \vec{V}(A/R_o)$$

3) Show that $2E_c(S/R_o)$ equals the product (Comoment) of the kinematic torsor and the kinetic torsor.

$$2E_c(S/R_o) = [T_V(S/R_o)]_A \otimes [T_C(S/R_o)]_A$$

where A is a point of the solid S.

Correction

Exercise 1:

A system (Σ) consists of a homogeneous rod OA of mass m and length L to which is articulated, at point A, a homogeneous disk of center $G \equiv A$, of mass M and radius R.

The rod OA rotates, in the plane $(O, \vec{x}_o, \vec{y}_o)$, around the axis $(O, \vec{z}_o)$. Its position is located by the angle ψ .

The disk rotates, in the plane $(O, \vec{x}_o, \vec{y}_o)$, around the extremity A of the rod. Its position with respect to R_o is located by the angle θ .

We denote by $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$ the laboratory reference frame, by $R_1(O, \vec{x}_1, \vec{y}_1, \vec{z}_1)$ the reference frame linked to the rod OA and $R_2(A, \vec{x}_2, \vec{y}_2, \vec{z}_2)$ a reference frame linked to the disk.

Determine:

- 1) The number of degrees of freedom of the system.
- 2) The kinetic resultant of the system Σ .
- 3) The angular momentum of the system Σ at point O with respect to the reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$.
- 4) The dynamic moment of the system Σ at point O with respect to the reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$.
- 5) The kinetic energy of the system Σ with respect to the reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$.

Correction

Exercise 2:

A perfect solid S consists of two glued parts: A disk S_1 of center O, of radius R and mass m, and a disk S_2 of center C ($OC = a$), of radius R/4 and same mass m. The solid S rotates at angular velocity ω around the axis $(O, \vec{z}_o)$ of the fixed reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$. The reference frame $R(O, \vec{x}, \vec{y}, \vec{z})$ is linked to the solid S.

- 1) Determine the position of the center of mass G of the solid S.
- 2) Calculate the angular momentum at O of the solid S. Deduce its moment of inertia I_{ZZ} with respect to the axis $(O, \vec{z}_o)$.
- 3) Find I_{ZZ} of the solid S by applying Huygens' theorem.
- 4) Calculate the angular momentum at G of the solid S.
- 5) Calculate the moment of inertia I_{GZ} of the solid S with respect to the axis $(G, \vec{z}_o)$ applying Huygens' theorem.
- 6) Calculate the kinetic energy $E_C(S/R_o)$ of the solid S.

Correction

Exercise 3:

A floor sanding machine is composed of an arm OAC of negligible mass such that $OA = L$, $AC = L/2$ and a disk of radius R and mass M. The arm is in rotational motion with respect to the fixed frame with a rotation speed $\dot{\psi} = \text{Cte}$. The disk rotates around the arm AC with a rotation speed $\dot{\theta} = \text{Cte}$. We will take the reference frame $R_1(O, \vec{x}_1, \vec{y}_1, \vec{z}_o = \vec{z}_1)$ as the projection reference frame.

Determine:

- 1) The instantaneous rotation velocity of the disk $\vec{\Omega}(D/R_o)$.
- 2) The absolute velocity and acceleration of point C.
- 3) The kinetic torsor of the disk at point O.
- 4) The dynamic torsor of the disk at point O.
- 5) The kinetic energy of the system with respect to R_o .

Correction

Exercise 4: (Homework)

Let the solid S be a homogeneous disk, of negligible section, of mass m, of center G and radius a (figure 3). We propose to study the motion of the solid S with respect to the fixed reference frame $R_o(O, \vec{x}_o, \vec{y}_o, \vec{z}_o)$. The axis $(O, \vec{z}_o)$ is the upward vertical.

We call $R(G, \vec{x}, \vec{y}, \vec{z})$ the direct orthonormal reference frame linked to the solid S.

A and C are its principal moments of inertia at G (C moment of inertia with respect to the axis $(G, \vec{z})$ axis of revolution symmetry of S). We assume throughout the problem that the solid S remains in contact with the horizontal plane (O, x_o, y_o) at point I. The vector $\vec{IG}$ is carried by the axis $(G, \vec{w})$. The solid S is located by the usual Euler angles ψ , θ and ϕ . It is in motion without slipping.

Express all results in the basis $(\vec{u}, \vec{w}, \vec{z})$.

- 1) Determine the instantaneous rotation velocity $\vec{\Omega}(S/R_o)$.
- 2) Determine the reduction elements at G of the kinetic torsor.
- 3) Determine the reduction elements at G of the dynamic torsor.
- 4) Calculate the kinetic energy of the solid with respect to R_0 .

Indication:

The condition of rolling without slipping gives:

$$\vec{V}_g(D/R_o) = \vec{V}(I \in D/R_o) = \vec{V}(G/R_o) + \vec{\Omega}(D/R_o) \times \overrightarrow{GI} = \vec{0}$$

Correction